

Sumit Barua

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RESEARCH FOCUS

Trustworthy, explainable, and uncertainty-aware AI for high-stakes healthcare and decision-support systems, with emphasis on multimodal healthcare AI, clinical/EMS decision support, model reliability under domain shift, human-AI collaboration, and grounded LLM/RAG systems for evidence-based reasoning.

EDUCATION

Western Michigan University, Kalamazoo, MI

Aug 2025 – Apr 2027

- M.S. in Computer Science
- GPA: 4.0 / 4.0
- Advisor: Prof. Alvis Fong
- Relevant Coursework: Generative AI, Machine Learning, Secured Programming, Cloud Computing, Information Retrieval Systems, MS Thesis

Western Michigan University, Kalamazoo, MI

Aug 2018 – Apr 2022

- B.S. in Computer Science
- GPA: 3.59 / 4.0
- Relevant Coursework: Java Programming, Data Structure and Algorithm, System Programming, Artificial Intelligence, Software Development

PUBLICATIONS

1. **Barua, S.**, Bahre, R., and Hong, G. Y. “Edge-Enabled Explainable Multimodal AI for Emergency Wound Assessment in Smart Ambulance IoMT Networks.” *IEEE Communications Magazine*, 2026, pp. 1–6. doi: 10.1109/MCOM.001.2500767.
2. **Barua, S.**, Bahre, R., Babiceanu, R., Hong, G. “Uncertainty-Aware Wound Localization and First-Aid Decision Support From Smartphone Images Using an Explainable Multimodal AI Framework.” *Cureus Journal of Computer Science*, vol. 3, May 18, 2026, es44389-026-00077-z. doi: 10.7759/s44389-026-00077-z.

MANUSCRIPTS UNDER REVIEW

1. **Barua, S.**, Jabran, S., Rodgers, C., Fong, A. “Quantifying Behavioral Contributions to Diabetes Risk: An Explainable Machine-Learning Analysis of BRFSS 2024.” *Accepted at Discover Public Health*.
2. **Barua, S.**, Hong, G.Y., Dursunoglu H., Rodgers, C., Fong, A. “CiteGuard-RAG: A Validation-Centered AI System for Evidence-Grounded Question Answering.” *Under review at IEEE Transactions on Artificial Intelligence*.
3. **Barua, S.**, Fong, A., Uddin, B. “Feature-Stable Explainable Machine Learning for IoMT Intrusion Detection Under Dataset Shift.” *Under review in Security & Privacy*.

SELECTED RESEARCH PROJECTS

Explainable Hybrid Multimodal Framework for Remote Wound Assessment [GitHub]

- Developed an explainable multimodal AI framework using *YOLOv11* for wound localization, *ResNet-50* for anatomical classification, and *Mistral-7B* LLM-assisted generation for first-aid decision support in smart-ambulance and remote-care settings.

- Integrated Grad-CAM-based visual explanation and bounding-box localization to support transparent wound assessment; resulted in two first-author publications in *IEEE Communications Magazine* and *Cureus*.

CiteGuard-RAG: Citation-Constrained System for Legal Question Answering [GitHub]

- Built a retrieval-augmented generation pipeline combining hybrid retrieval, citation-constrained generation, sentence-level validation, and bounded regeneration using *Mistral-7B* and *Sentence Transformers*.
- Evaluated reliability under controlled and external domain-shift settings, achieving 95.4% grounded answer accuracy, 100% citation validity, and a 1.5% hallucination rate.

Explainable Machine Learning for Diabetes Risk Analysis Using BRFSS 2024 [GitHub]

- Trained interpretable machine-learning models on BRFSS 2024 data to classify no diabetes, prediabetes, and diabetes risk using behavioral, demographic, and anthropometric predictors.
- Applied SHAP-based explainability to quantify behavioral contributions to diabetes risk and support transparent public-health decision analysis.

Robust IoMT Intrusion Detection Under Cross-Dataset Shift [GitHub]

- Designed explainable machine-learning models for IoMT intrusion detection using network-traffic behavior features under cross-attack-family and cross-dataset shift conditions.
- Evaluated robustness, feature stability, and SHAP-based behavioral explanations to support trustworthy cyber-resilience in connected healthcare infrastructure.

PROFESSIONAL EXPERIENCE

Graduate Research Assistant

Jan 2026 – Present

WMU Department of Computer Science

Kalamazoo, MI

- Develop machine-learning and explainable-AI workflows for research problems involving IoMT systems, smart ambulance networks, healthcare decision support, and cyber-resilient connected systems.
- Develop and maintain Unity-based interactive simulation environments for an NSF-funded research project, supporting experimental workflows, user interaction, and applied AI/game-based system development.
- Contribute to manuscript development, experiment design, dataset preprocessing, model evaluation, and technical documentation for peer-reviewed research outputs.

Student IT Assistant

Aug 2025 – Dec 2025

WMU Office of Information Technology

Kalamazoo, MI

- Provide technical support to students and faculty for computing systems, software, hardware, and network-related issues.
- Support troubleshooting of classroom and network systems, strengthening practical understanding of real-world computing environments.

Software Engineer

Jun 2022 – May 2025

Dematic Corporation

Grand Rapids, MI

- Developed automation software for order-fulfillment and material-handling systems.
- Supported system integration, testing, troubleshooting, and deployment of reliable industrial automation software.
- Collaborated with cross-functional engineering teams to deliver practical software solutions for warehouse automation environments.

Student Information Tech Assistant
WMU College of Health & Human Services

Dec 2019 – Apr 2022
Kalamazoo, MI

- Answer user inquiries regarding computer software or hardware operation to resolve problems.
- Install and perform minor repairs to hardware, software, or peripheral equipment, following design or installation specifications.

Software Engineering Intern
Frontier Communications

May 2021 – Aug 2021
Allen, TX

- Designed and implemented a web application using Flask and Jinja to fetch the network port information and their real-time status.
- Developed tools to provide automated basic troubleshooting based on the port status for speedy customer service, which reduces the time for on-call troubleshooting for the customer service team.

Undergraduate Learning Assistant
WMU Student Success Center

Aug 2019 – Dec 2019
Kalamazoo, MI

- Supported students in learning Java programming fundamentals, object-oriented programming, algorithmic thinking, and problem-solving techniques.
- Guided students through programming exercises and debugging, helping them identify syntax, runtime, and logical errors.

TECHNICAL SKILLS

Programming:	Python, SQL, Java, C
Machine Learning / AI:	scikit-learn, TensorFlow, XGBoost, YOLOv11, ResNet-50, SHAP, Grad-CAM, Retrieval-Augmented Generation, Large Language Models
Research Domains:	Trustworthy AI, Explainable AI, Healthcare AI, IoMT, Edge AI, Cybersecurity, Cyber-Physical Systems, Vehicular Networks, Intelligent Transportation Systems
Development Tools:	Git, Linux, Django, Flask, Streamlit, REST APIs, MS SQL Server, Unity
Research Methods:	Dataset preprocessing, feature engineering, model evaluation, explainability analysis, domain-shift evaluation, uncertainty-aware decision support

HONORS AND AWARDS

- WMU Global Education Graduate Merit Scholar 2025 – 2027, Western Michigan University
- WMU Haenicke Global Education Undergraduate Merit Scholar 2018 – 2022, Western Michigan University.